





Practice Exam 2024

1.1

To calculate MLE of w , we use a following equation

$$w_{MLE} = (\Phi^T \Phi)^{-1} \Phi^T y, \text{ where } \Phi = [1, x^2, \sin(x), x]$$

- I obtain:

$$w_{MLE} = \begin{bmatrix} -0.73 \\ 0.11 \\ 2.35 \\ 1.01 \end{bmatrix}$$

To calculate σ^2_{MLE} we use $\frac{1}{N} \sum_{i=1}^N (y_i - w_{MLE}^T \phi(x_i))^2$

I obtain $\sigma^2_{MLE} = 0.26$, to obtain β , we need $\beta = \frac{1}{\sigma^2} = 4.84$

1.2

To calculate the post. pred. dist, we use the $x^* = 1$ and plug it into our design matrix. Then the plug in approx is:

$$P(y^* | y, x^* = 1) \approx \mathcal{N}(\phi_*^T w_{MLE}, \sigma^2)$$

- since we have a plug in approx we ignore uncertainty in σ^2 .

$$\therefore P(y^* | y, x^* = 1) \approx \mathcal{N}(2.37, 0.21)$$

$$\therefore \mu_{y^*} = 2.37, \sigma = 0.48, 95\% \text{ CI} = [1.48, 3.26],$$

Part 2

Suppose the outcome of $N = 31$ independent Bernoulli trials generated $y = 7$ successes. Let $\theta \in [0, 1]$ denote the probability of success. Assume a Binomial likelihood, i.e. $p(y|\theta) = \text{Bin}(y|N, \theta)$ with the following prior distribution for θ :

$$p(\theta) = \frac{3}{7} \text{Beta}(\theta|2, 10) + \frac{4}{7} \text{Beta}(\theta|10, 2) \quad (3)$$

Question 2.1: Compute the prior probability of the event $\theta > \frac{1}{2}$.

$$\begin{aligned} P(\theta > 0.5) &= \int_{0.5}^{1.0} p(\theta) d\theta \\ &= \int_{0.5}^{1.0} \left[\frac{3}{7} \text{Beta}(\theta|2, 10) + \frac{4}{7} \text{Beta}(\theta|10, 2) \right] d\theta \\ &= \frac{3}{7} \left[1 - \int_0^{0.5} \frac{3}{7} \text{Beta}(\theta|2, 10) d\theta \right] + \frac{4}{7} \left[1 - \int_0^{0.5} \frac{4}{7} \text{Beta}(\theta|10, 2) d\theta \right] \end{aligned}$$

Question 2.2: Compute the analytical marginal likelihood $p(y)$ and evaluate $p(y = 7)$.

- Likelihood: $y|\theta \sim \text{Bin}(N, \theta)$
- Prior: $p(\theta) = \frac{3}{7} \text{Beta}(\theta|2, 10) + \frac{4}{7} \text{Beta}(\theta|10, 2)$
- We want: $p(y) = \int_0^1 p(y|\theta) p(\theta) d\theta$

$$\begin{aligned} p(y) &= \int p(y|\theta) \left[\frac{3}{7} \text{Beta}(\theta|2, 10) + \frac{4}{7} \text{Beta}(\theta|10, 2) \right] d\theta \\ &= \frac{3}{7} \int p(y|\theta) \text{Beta}(\theta|2, 10) d\theta + \frac{4}{7} \int p(y|\theta) \text{Beta}(\theta|10, 2) d\theta \end{aligned}$$

- due to linearity, we split the integrals into two

Write the Binom. lik and Beta prior in pdf form

- $p(y|\theta) = \binom{N}{y} \theta^y (1-\theta)^{N-y}$
- $p(\theta|\alpha, \beta) = \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)}$, $0 < \theta < 1$

$$B(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

- multiply and integrate over $\theta \in [0, 1]$

$$\begin{aligned} \int p(y|\theta) \text{Beta}(\theta|\alpha, \beta) d\theta &= \binom{N}{y} \frac{1}{B(\alpha, \beta)} \int_0^1 \theta^{y+\alpha-1} (1-\theta)^{N-y+\beta-1} d\theta \\ &= \binom{N}{y} \frac{B(y+\alpha, N-y+\beta)}{B(\alpha, \beta)} \end{aligned}$$

Apply this to two Betas and combine.

For the first $\text{Beta}(\alpha, \beta) = (2, 10)$

$$\binom{N}{y} \cdot \frac{B(y+2, N-y+10)}{B(2, 10)}$$

For the second $\text{Beta}(\alpha, \beta) = (10, 2)$

$$\binom{N}{y} \cdot \frac{B(y+10, N-y+2)}{B(10, 2)}$$

- combine the mixture weights

$$p(y) = \frac{3}{7} \binom{N}{y} \cdot \frac{B(y+2, N-y+10)}{B(2, 10)} + \frac{4}{7} \binom{N}{y} \cdot \frac{B(y+10, N-y+2)}{B(10, 2)}$$

Plug in the numbers, $N=31$, $y=7$

$$p(y=7) = \frac{3}{7} \binom{31}{7} \cdot \frac{B(9, 24)}{B(2, 10)} + \frac{4}{7} \binom{31}{7} \cdot \frac{B(17, 26)}{B(10, 2)} \approx \underline{\underline{0.03}}$$

$$\begin{aligned}
 \log p(y_n | w, x_n) p(w) &= \sum_{n=1}^N \log p(y_n | w, x_n) + \log p(w) \\
 &= -\cancel{\frac{1}{2} \log(2\pi)} - \frac{1}{2} \alpha w^T w + \sum \left[y_n \lambda_n + (-\lambda_n) + \log \frac{1}{y_n!} \right] \\
 &= -\frac{1}{2} \alpha w^T w + \sum \left[y_n \log(\lambda_n) - \lambda_n \log \right] + K \\
 &= -\frac{1}{2} \alpha w^T w + \sum_{n=1}^N y_n \log \lambda_n - \sum_{n=1}^N \lambda_n + K
 \end{aligned}$$

$$\underbrace{L(w)}_{\substack{\text{log posterior up} \\ \text{to a constant}}} = -\frac{\alpha}{2} w^T w + \sum_n y_n \underbrace{(x_n^T w)}_{\eta_n} - \sum_{n=1}^N \exp\left(\frac{x_n^T w}{\lambda_n}\right) + K$$

- $w \in \mathbb{R}^d$, $X \in \mathbb{R}^{N \times d}$, $y \in \mathbb{R}^N$
- $\eta = Xw \in \mathbb{R}^N$, $\lambda = \exp(\eta) \in \mathbb{R}^N$
- $w^T w \Rightarrow \text{inp. dot}(w, w)$, w shape $(d,)$
- $\eta = Xw \Rightarrow \text{eta} = X @ w$, where X is (N, d)

⚠ keep w 1D $(d,)$, so that $X @ w$ becomes $(N,)$ and **NOT** $(d,)$

5.5

Goal: mean and variance of the predictive distribution:

$$P(y^* | y, x^* = 0) = \int \text{Poisson}(y^* | \underbrace{\lambda^*(\omega^*)}_{\text{likelihood at } x^*}) \underbrace{p(\omega | y)}_{\text{posterior over } \omega} d\omega$$

$$\lambda^*(\omega) = \exp(\tilde{x}^T \omega) = \exp(\omega_0 + \omega_1 x^*), \quad \tilde{x} = \begin{bmatrix} 1 \\ x^* \end{bmatrix}$$

- we have the MAP and Hessian of the log posterior

$$W_{\text{MAP}} = \begin{bmatrix} 2.1575 & -0.5201 \end{bmatrix}^T$$

$$H = \begin{bmatrix} -9 & -17 \\ -17 & -48 \end{bmatrix} \Rightarrow S = -H^{-1}$$

- we draw many samples from the posterior $\omega^{(i)} \sim \mathcal{N}(\omega | W_{\text{MAP}}, S)$ and then pass them through the model to get rates λ^* then draw predictive counts y^* . Sample average approx exp

From Laplace we have $\omega | y \approx \mathcal{N}(\mu, \Sigma)$, $\mu = W_{\text{MAP}}$, $\Sigma = (-H)^{-1}$

$\forall \omega^{(i)}$

1. Linear predictor: $\eta^{(i)} = \tilde{x}^T \omega^{(i)}$

2. Rate: $\lambda^{(i)} = \exp(\eta^{(i)})$

3. Predictive count: $y^{(i)} \sim \text{Poisson}(\lambda^{(i)})$

Then $\mathbb{E}[y^* | y, x^*] \approx \frac{1}{S} \sum_{i=1}^S y^{(i)}$, $\text{Var}(y^* | y, x^*) \approx \text{sample variance}(y^{(i)})$